UTKARSH SINGH

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Professional Summary

Undergrad AI researcher specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning. I care about taste in engineering. From hackathon floors to research labs, I ship fast, break hard problems into working prototypes, and care deeply about systems that solve real user pain.

Education

Maharaja Agrasen Institute of Technology [GGSIPU] — New Delhi, India
B.Tech in Computer Science Engineering, Specialization in Artificial Intelligence (9.3 CGPA) — Aug. 2025 - May 2029

Kamal Public Sr. Sec. School [CBSE] — New Delhi, India
Class XII PCM with Computer Science (92.4%) — April 2011 - May 2025

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C++, C

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, PyTorch, TensorFlow, Transformers, Hugging Face, scikit-learn

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs

Frontend: React, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, SQLite

MLOps & Cloud: Docker, Git, GitHub Actions (CI/CD), AWS (S3, EC2, Lambda), GCP, Vercel, Render, Railway
Libraries & Tools: NumPy, Pandas, LangChain, Matplotlib, OpenCV, Postman, Jupyter

Projects

KAVACH: Fraud Risk Manager for Indian D2C Merchants

- Python, FastAPI, XGBoost, SHAP, NetworkX, Next.js: built a calibrated return-risk scorer paired with graph-based fraud ring detection over shared phones, devices, VPAs and fuzzy-matched addresses.
- Defence-only risk manager for Indian D2C merchants targeting COD return abuse and coordinated chargebacks that erode 5-15% of gross margin.
- Ring detection scored 0.889 precision at 1.000 recall on held-out data; every flagged order ships with SHAP attributions and a SHA-256 hash-chained audit trail.

NETRA: AI Growth Copilot for Indian Kirana Stores

- Python, FastAPI, Llama 3.3 70B, Sarvam AI, React, Differential Privacy: built demand forecasting, non-collusive price benchmarking and credit scoring delivered via Soundbox audio, WhatsApp and web in 7 Indic languages.
- AI growth copilot for Indian Kirana stores serving 90% of a $600B+ grocery market, enabling solo merchants to make data-backed pricing and inventory decisions.
- Enforces a five-layer privacy boundary using k-anonymity cohort gating and Laplace DP to block competitor surveillance and prevent price collusion.

ATLASS: Adaptive Taste Learning And Suggestion System

- JavaScript, SVD, LSA/TF-IDF, WebGL, GSAP, TMDb API: engineered SVD fold-in blended 70/30 with a 48-dimensional LSA content profile, pre-trained weights shipped as ∼700KB JSON, re-scoring in under 50ms.
- Hybrid movie recommender running entirely in-browser with zero inference server, built on MovieLens (9,742 films, 100K+ ratings).
- WebGL gallery renders recommendation cards along a parametric arc with GSAP transitions for an interactive browsing experience.

Achievements

Top 10 Finalist, Paytm Build for India AI Hackathon — 2026

- Finished Top 10 nationally; built NETRA, an AI growth copilot for Indian merchants with privacy-preserving market intelligence using k-anonymity and differential privacy.

Accepted Research Paper at ICML-Scale (International Conference on Machine Learning - Scale) 2026

- Co-authored **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding**, accepted at ICML-Scale 2026 as a second-year undergraduate.

Honoured, Award of Excellence — 2025

- Honored with the Academic Excellence Award, presented by Shri Suresh Jain (National Organization Secretary, Bharat Vikas Parishad) and Shri Vijay Kumar (Joint Commissioner of Police).

Finalist, ET Gen AI Hackathon — 2026

- Co-built an Autonomous Multi-Agent AI Code Intelligence & Remediation Platform; ranked Top Performing out of 3000+ candidates nationwide.